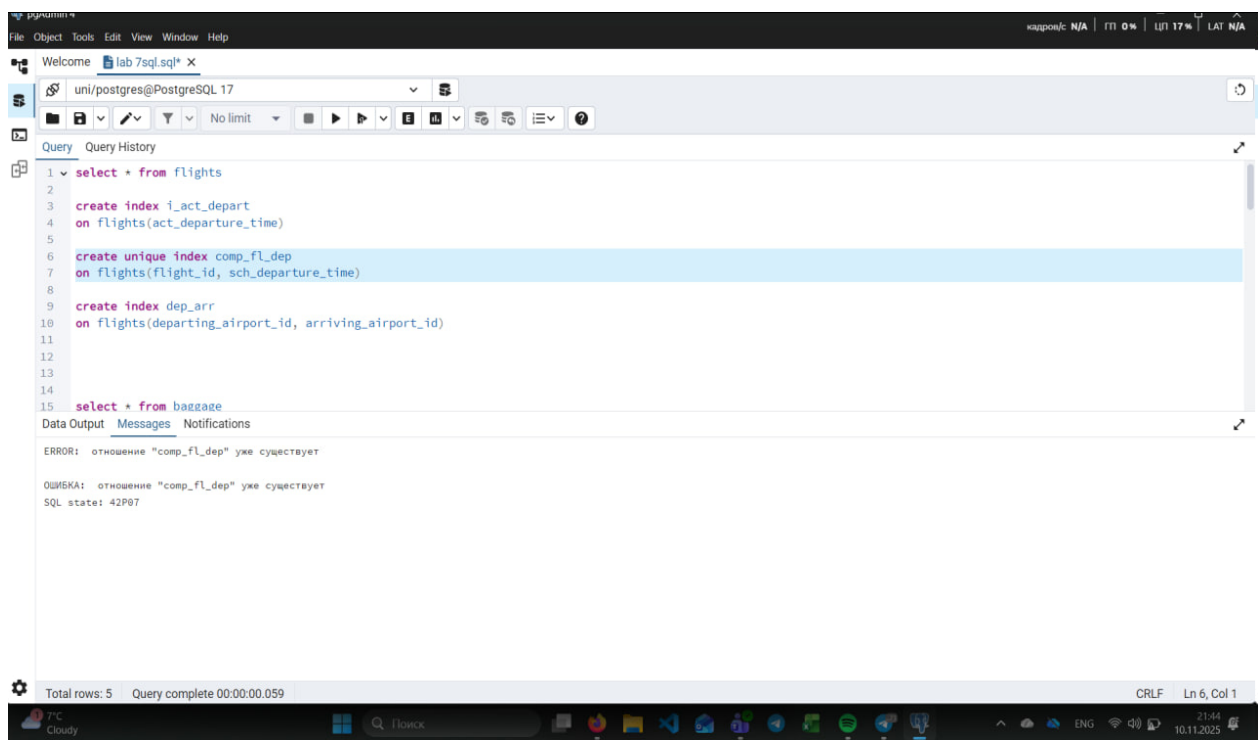
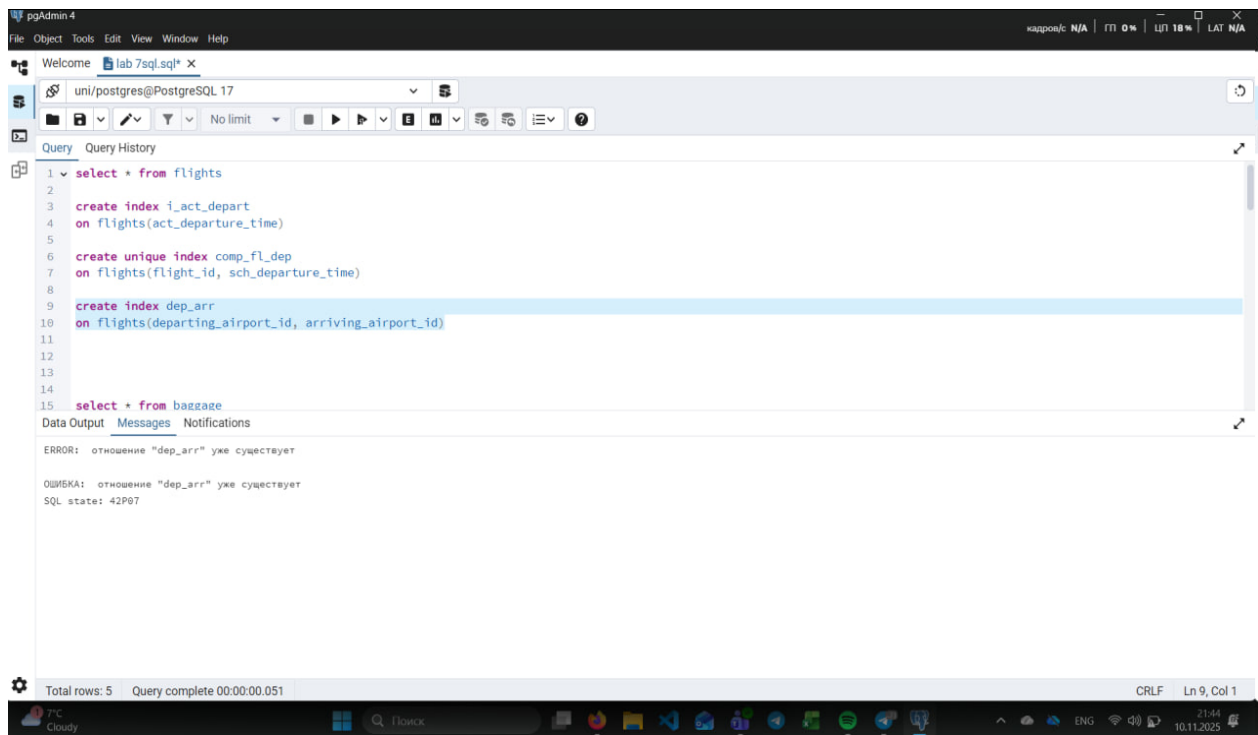




pgAdmin 4

File Object Tools Edit View Window Help

Welcome lab 7sql.sql X

uni/postgres@PostgreSQL 17

No limit

Query Query History

```

1 select * from flights
2
3 create index i_act_depart
4 on flights(act_departure_time)
5
6 create unique index comp_fl_dep
7 on flights(flight_id, sch_departure_time)
8
9 create index dep_arr
10 on flights(departing_airport_id, arriving_airport_id)
11
12
13
14
15 select * from baggage

```

Data Output Messages Notifications

ERROR: отношение "i_act_depart" уже существует

ОШИБКА: отношение "i_act_depart" уже существует

SQL state: 42P07

Total rows: 5 Query complete 00:00:00.048 CRLF Ln 3, Col 1

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File Object Tools Edit View Window Help

Welcome lab 7sql.sql X

uni/postgres@PostgreSQL 17

No limit

Query Query History

```

17
18 explain analyze select first_name from passengers
19
20 select * from baggage
21 explain analyze select weight_in_kg from baggage where weight_in_kg > 10
22
23 create index weigths_10
24 on baggage(weight_in_kg)
25 where weight_in_kg > 10;
26
27 analyze baggage
28 explain analyze select weight_in_kg from baggage where weight_in_kg > 20
29
30 explain analyze select * from flights
31 where departing_airport_id = 5 and arriving_airport_id = 3;

```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

QUERY PLAN	
text	
1	Seq Scan on baggage (cost=0.00..3.55 rows=95 width=5) (actual time=0.025..0.041 rows=96 loops=...
2	Filter: (weight_in_kg > 20::numeric)
3	Rows Removed by Filter: 28
4	Planning Time: 0.384 ms
5	Execution Time: 0.057 ms

Successfully run. Total query runtime: 62 msec. 5 rows affected. X

Total rows: 5 Query complete 00:00:00.062 CRLF Ln 27, Col 1

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File Object Tools Edit View Window Help

Welcome lab 7sql.sql x

uni/postgres@PostgreSQL 17

Query Query History

```

23 create index weights_10
24 on baggage(weight_in_kg)
25 where weight_in_kg > 10;
26 analyze baggage
27 explain analyze select weight_in_kg from baggage where weight_in_kg > 20
28
29
30 explain analyze select * from flights
31 where departing_airport_id = 5 and arriving_airport_id = 3;
32
33
34 select * from passengers
35
36 delete from passengers p
37 USING passengers q

```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

QUERY PLAN
1 Index Scan using dep_arr on flights (cost=0.14..8.16 rows=1 width=72) (actual time=0.019..0.020 rows=0 loops=...)
2 Index Cond: ((departing_airport_id = 5) AND (arriving_airport_id = 3))
3 Planning Time: 0.164 ms
4 Execution Time: 0.055 ms

Total rows: 4 Query complete 00:00:00.060 CRLF Ln 30, Col 1

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File Object Tools Edit View Window Help

Welcome lab 7sql.sql x

uni/postgres@PostgreSQL 17

Query Query History

```

42 -- passengers (passport_number)
43
44 INSERT INTO passengers
45 (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, updated_at)
46 VALUES
47 (11, 'ilyas', 'makaka', '1943-06-22', 'male', 'Kazakhstan', 'Japan', 'D2233601', '2022-06-19 11:28:12', '2022-06-16 08:49:42');
48 INSERT INTO passengers
49 (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, updated_at)
50 VALUES
51 (12, 'elsa', 'makaka', '1943-06-22', 'female', 'Kazakhstan', 'Japan', 'D2233611', '2022-06-19 11:28:12', '2022-06-16 08:49:42');
52
53 select count(distinct(passport_number)), count(passport_number) from passengers
54 --- so because there were only 5 real distinct passport_numbers, our unique index couldn't pass, so i decided simply to delete all of this (on delete cascade)
55 --- and now i have 2 more rows, that fit the unique constraint, and i have an index ready to work
56

```

Data Output Messages Notifications

DROP INDEX

Query returned successfully in 73 msec.

Total rows: Query complete 00:00:00.073 CRLF Ln 27, Col 1

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File Object Tools Edit View Window Help

Welcome lab 7sql.sql x

uni/postgres@PostgreSQL 17

Query Query History

```

62 select * from passengers
63
64 explain analyze select first_name, last_name, date_of_birth, country_of_citizenship from passengers
65 where extract(year from date_of_birth) = 1984 and country_of_citizenship = 'Philippines'
66
67 --- no the query didn't used indexes, because we have little amount of data, and because i didn't create index with extract(year from date_of_birth)
68 --- postgresql used seq scan, and not the index one
69
70
71
72 select indexname, indexdef
73 from pg_indexes
74 where tablename = 'passengers'
75
76 drop index first_name_index, pass_idx_uni, person_check_b

```

Data Output Messages Notifications

DROP INDEX

Query returned successfully in 73 msec.

Total rows: Query complete 00:00:00.073 CRLF Ln 76, 27

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File Object Tools Edit View Window Help

Welcome lab 7sql.sql x

uni/postgres@PostgreSQL 17

Query Query History

```

54 --- so because there were only 5 real distinct passport_numbers, our unique index couldn't pass, so i decided simply to delete all of this (on delete cascade)
55 --- and now i have 2 more rows, that fit the unique constraint, and i have an index ready to work
56
57
58
59
60 create index person_check_b
61 on passengers (country_of_citizenship, date_of_birth, last_name, first_name);
62 select * from passengers
63
64 explain analyze select first_name, last_name, date_of_birth, country_of_citizenship from passengers
65 where extract(year from date_of_birth) = 1984 and country_of_citizenship = 'Philippines'
66
67 --- no the query didn't used indexes, because we have little amount of data, and because i didn't create index with extract(year from date_of_birth)
68 --- postgresql used seq scan, and not the index one

```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

QUERY PLAN	
	text
1	Seq Scan on passengers (cost=0.00..3.12 rows=1 width=27) (actual time=0.014..0.015 rows=0 loops=1)
2	Filter: (((country_of_citizenship)::text = 'Philippines'::text) AND (EXTRACT(year FROM date_of_birth) = '1984'::numeri...
3	Rows Removed by Filter: 7
4	Planning Time: 1.553 ms
5	Execution Time: 0.027 ms

Total rows: 5 Query complete 00:00:00.054 CRLF Ln 68, 27

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